

Part I

Trade, Conflict, and Security

Parts I–VI assumed that the international system is *lawful*: property is secure, endowments are given, and the only instruments in play are taxes on voluntary exchange. This part drops that assumption. Sovereign states cannot sign complete, enforceable long-term contracts with one another; they arm to defend or extend claims on resources, and trade takes place in an essentially **anarchic** environment. Two grand traditions collide here. The **classical liberal** view holds that trade not only raises welfare but pacifies: interdependence makes war unthinkable. The **realist** view replies that trade generates a *security externality* (Gowa): by changing the value of contested resources and the opportunity cost of arming, openness can *intensify* conflict — which is why states have always used embargoes, sanctions, and export controls as extensions of security policy. The Skaperdas–Syropoulos–Garfinkel research program formalizes both views inside one framework and asks: *when does each hold?* The organizing trick throughout is to make **guns** a choice variable — labor or resources diverted from producing “butter” to enforcing claims — and to let the trade regime determine the marginal benefit and opportunity cost of guns. Lessons 14–15 build the two foundational models (the attached AER P&P and Economic Journal papers); Lessons 16–19 develop the starred core of the reading list (insecure resources in general equilibrium, trade with enemies and friends, self-enforcing peace, prudence versus predation); Lesson 20 surveys the 2023–25 frontier (war-and-peace dynamics with trade, geopolitics of technology transfer, and the theory of sanctions).

14 Lesson 14: Guns, Butter, and Openness — Skaperdas–Syropoulos (2001)

Skaperdas, S. and C. Syropoulos (2001), “Guns, Butter, and Openness: On the Relationship between Security and Trade,” *AER Papers & Proceedings*, 91(2), 353–357

Question. Does opening to trade reduce or increase arming between two countries that dispute a resource? **Setting.** Two *small* countries contest land (oil); labor produces guns or butter one-to-one; Cobb–Douglas preferences. **Headline results.** (i) Trade *levels the playing field* for arming: it equalizes the opportunity cost of guns across countries, so arming under trade is unrelated to arming under autarky. (ii) If the world price of butter exceeds its autarky level, opening to trade *reduces* arming and raises welfare beyond the ordinary gains from trade (the liberal view). (iii) For an intermediate range of world prices *below* the autarky price, the negative security externality of trade outweighs the gains from trade and **autarky is superior** (the realist view). (iv) Trade intervention can be welfare-improving if, by precommitment, it induces the adversary to arm less.

14.1 Setup and the indirect utility function

Country $i = 1, 2$ owns T_i units of secure land and L_i units of secure labor; T_0 units of land are *disputed*. One unit of land yields one unit of oil; one unit of labor yields one gun or one unit of butter. Preferences are Cobb–Douglas,

$$U_i(O_i, B_i) = O_i^\alpha B_i^\beta, \quad \alpha + \beta = 1.$$

Claims on T_0 are established by **guns** through the contest success function (CSF)

$$\phi_i = \frac{G_i^m}{G_1^m + G_2^m}, \quad m \leq 1,$$

so country i controls land $T_i + \phi_i T_0$ and has butter-capable labor $L_i - G_i$. With p_i the relative price of butter (oil the numeraire), GDP is

$$R_i = T_i + \phi_i T_0 + p_i(L_i - G_i).$$

Maximizing U_i subject to $O_i + p_i B_i = R_i$ gives the Cobb–Douglas demands $O_i = \alpha R_i$ and $p_i B_i = \beta R_i$, hence the **indirect utility function**

$$V_i = \Gamma p_i^{-\beta} R_i, \quad \Gamma \equiv \alpha^\alpha \beta^\beta.$$

Guns affect welfare through *two channels*: they raise R_i via $\phi_i T_0$ but lower it via forgone butter labor; and (under autarky) they move the domestic price p_i .

14.2 Autarky: derivation of equilibrium arming

Under autarky the butter market clears domestically: supply $L_i - G_i$ equals demand $\beta R_i / p_i$. Combining with $O_i = \alpha R_i = T_i + \phi_i T_0$ yields the market-clearing price

$$p_i^A = \frac{\beta}{\alpha} \cdot \frac{T_i + \phi_i T_0}{L_i - G_i},$$

increasing in own guns (more oil supplied, less butter). Substituting p_i^A and $R_i = (T_i + \phi_i T_0) / \alpha$ into V_i collapses autarky welfare to

$$V_i^A \propto (T_i + \phi_i T_0)^\alpha (L_i - G_i)^\beta$$

— a Cobb–Douglas in the two things guns move in opposite directions. Taking logs and differentiating, the FOC is

$$\frac{\alpha (\partial \phi_i / \partial G_i) T_0}{T_i + \phi_i T_0} = \frac{\beta}{L_i - G_i}.$$

In the symmetric case ($T_i = T$, $L_i = L$, so $\phi_i = \frac{1}{2}$ and $\partial \phi_i / \partial G_i = m / 4G$), solving for G :

Equilibrium arming under autarky

$$G^A = \frac{\frac{m}{4}}{\frac{m}{4} + \frac{\beta}{\alpha} \left(\frac{1}{2} + t \right)} L, \quad t \equiv T / T_0,$$

increasing in labor L and conflict effectiveness m ; decreasing in land security t and in the taste for butter β / α . Substituting back, the equilibrium autarky price is

$$\tilde{p}^A = \frac{T_0}{L} \left[\frac{m}{4} + \frac{\beta}{\alpha} \left(\frac{1}{2} + t \right) \right].$$

Everything about domestic arming — endowments, security, tastes — is priced into $\tilde{p}^A$, the benchmark against which the world price will be compared.

14.3 Free trade: leveling the playing field

Under free trade both (small) countries face the fixed world price p ; the indirect utility is $V_i^F = \Gamma p^{-\beta} R_i$ and the FOC no longer involves the price channel:

$$\frac{\partial R_i}{\partial G_i} = \underbrace{\frac{\partial \phi_i}{\partial G_i} T_0}_{\text{marginal claim on oil}} - \underbrace{p}_{\text{opportunity cost}} = 0 \quad \Rightarrow \quad \boxed{G^F = \frac{mT_0}{4p}}$$

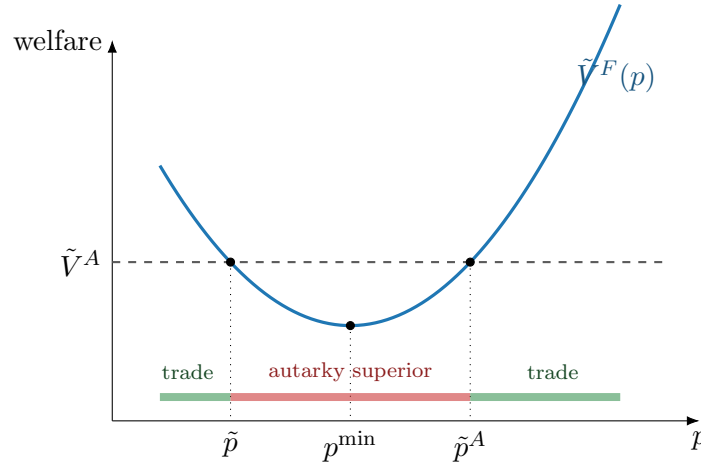
(interior, i.e. when the labor constraint $G \leq L$ does not bind).

Trade levels the playing field for arming

Under trade both countries face the *same* opportunity cost of guns, p , so they arm identically *regardless* of their labor endowments, secure land, or tastes — all of which mattered under autarky. G^F depends only on the size of the prize T_0 , conflict effectiveness m , and the world price p : a *higher* world price of butter makes guns dearer and cuts arming; a *lower* one makes guns cheap and fuels arming. Arming under trade and arming under autarky are therefore essentially unrelated, and trade can bring either much more or much less arming than autarky.

14.4 Trade versus autarky: the welfare comparison

Equilibrium free-trade welfare $\tilde{V}^F(p)$ is *convex* in p , with three critical prices: its minimizer $p^{\min}$; the autarky price $\tilde{p}^A$, at which $\tilde{V}^F(\tilde{p}^A) = \tilde{V}^A$ (opening at the autarky price changes nothing directly); and a lower threshold $\tilde{p} < p^{\min}$ at which $\tilde{V}^F(\tilde{p}) = \tilde{V}^A$ again.



Result: when the liberal view holds, and when the realist view does

$$\tilde{V}^F(p) \leq \tilde{V}^A \iff p \in [\tilde{p}, \tilde{p}^A].$$

$p > \tilde{p}^A$ (**liberal region**). Butter is more valuable on world markets than at home; the opportunity cost of guns rises; both adversaries arm *less*; trade brings the ordinary gains *plus* a peace dividend. $\tilde{p} < p < \tilde{p}^A$ (**realist region**). Cheap butter (valuable oil) makes guns cheap; both arm *more*; the negative security externality swamps the gains from trade. $p < \tilde{p}$. Arming is capped (the rival ends up specialized in guns, $G_j = L$), the externality is

exhausted, and the ever-improving terms of trade eventually make trade beneficial again.

To see the mechanics, differentiate equilibrium free-trade welfare using the envelope theorem (own guns are optimized, so only the direct price effect and the *rival's* response survive):

$$\frac{d\tilde{V}_i^F}{dp} = \Gamma p^{-\beta} \left[\underbrace{-M_i^F}_{\text{terms-of-trade effect}} + \underbrace{T_0 \frac{\partial \phi_i}{\partial G_j} \frac{dG_j^F}{dp}}_{\text{security externality}} \right], \quad \frac{\partial \phi_i}{\partial G_j} < 0, \quad \frac{dG_j^F}{dp} < 0,$$

where M_i^F is the excess demand for butter. Starting exactly at $p = \tilde{p}^A$ (so $M_i^F = 0$), only the externality operates: a price *rise* disarms the rival and raises welfare unambiguously; a price *fall* arms the rival and lowers it. This is the formal content of the “security externality of trade,” and it is the master equation for Lessons 16–17.

Policy corollary: security-motivated trade intervention

With secure property, a small country's optimal policy is free trade (Lesson 1). With an *insecure* resource, trade intervention can raise welfare *if* it induces the adversary to produce fewer guns — but this requires **precommitment**: the trade regime must be locked in before security policies are chosen. Embargoes and sanctions are thus interpretable as security policy, not (failed) trade policy.

Understanding Quiz 14 — Lesson 14

- Q14.1** Under free trade, equilibrium arming $G^F = mT_0/4p$ depends on:
- (A) labor endowments and tastes, as under autarky
 - (B) only the prize T_0 , conflict effectiveness m , and the world price p — trade equalizes the opportunity cost of guns across the adversaries
 - (C) the secure land endowments T_i
 - (D) nothing — it is constant
- Q14.2** Autarky is superior to free trade exactly when:
- (A) never — small countries always gain from trade
 - (B) the world price of butter lies in the intermediate interval $(\tilde{p}, \tilde{p}^A)$, where the negative security externality outweighs the gains from trade
 - (C) the world price exceeds the autarky price
 - (D) the country exports oil
- Q14.3** Starting at $p = \tilde{p}^A$, a small *increase* in the world price of butter raises welfare because:
- (A) the country's terms of trade improve (it trades at $\tilde{p}^A$)
 - (B) excess demand is zero, so only the security externality operates: the rival's opportunity cost of guns rises and it disarms
 - (C) guns become cheaper
 - (D) oil becomes more valuable
- Q14.4** Trade intervention can be welfare-improving in this model only if:
- (A) the country is large in world markets
 - (B) it is precommitted before security policies are chosen, so that it induces the adversary to arm less
 - (C) tariff revenue is redistributed

(D) demand is linear

15 Lesson 15: Insecure Property and the Efficiency of Exchange — Skaperdas–Syropoulos (2002)

Skaperdas, S. and C. Syropoulos (2002), “Insecure Property and the Efficiency of Exchange,” *Economic Journal*, 112(476), 133–146

Question. Is *ex post* efficient exchange always *ex ante* Pareto optimal when property is insecure and enforcement (arming) costs are endogenous? **Answer: no.** The *expectation* of full tradeability enlarges the bargaining set — not just the contested resource but the entire output becomes negotiable — which intensifies arming. A more *limited* settlement (divide only the contested resource, then go separate ways) can be *ex ante* superior for one agent or even Pareto dominant. Only when the gains from trade are large enough is unlimited exchange *ex ante* efficient. The result rationalizes a first-order historical fact: the removal of restrictions on exchange and the development of *secure property rights* have gone hand in hand (Britain; feudal autarky; mercantilist market denial; North’s enforcement costs).

15.1 The framework

Two agents contest land. Agent i owns T_i secure units of land, R_i units of inalienable labor; T_0 units of land are insecure. Guns G_i leave productive labor $L_i = R_i - G_i$. Payoffs come from a neoclassical function $F(T, L)$ that is *twice differentiable, increasing, strictly concave, and linearly homogeneous* (C1) — read it as a production function (land $\rightarrow$ oil, labor $\rightarrow$ butter) or directly as utility. Winning probabilities $p(G_1, G_2)$ satisfy the standard CSF conditions (C2): symmetry, own-gun monotonicity, concavity. Under **winner-take-all conflict**, expected payoffs are

$$U^1 = p F(T_1 + T_0, R_1 - G_1) + (1 - p) F(T_1, R_1 - G_1),$$

and symmetrically for agent 2.

Given guns, settlement dominates conflict

By strict concavity of F in land,

$$F(T_1 + pT_0, L_1) = F[p(T_1 + T_0) + (1-p)T_1, L_1] > p F(T_1 + T_0, L_1) + (1-p) F(T_1, L_1) = U^1,$$

and likewise for agent 2: **dividing T_0 in proportion to the winning probabilities Pareto-dominates fighting** — conflict is a lottery over land, and concavity makes the sure share worth more than the gamble. Conflict payoffs nevertheless matter as the **disagreement point** of every negotiation: guns are chosen first, irreversibly and non-cooperatively (as in incomplete-contract theory, arming is a non-contractible investment), and only then do the parties bargain.

15.2 Two regimes: Limited Settlement versus Exchange

Under **Limited Settlement** the agents divide only the contested land (share γ to agent 1, determined by Nash bargaining from the conflict threat point) and then autarkically consume:

$$V^1 = F(T_1 + \gamma T_0, R_1 - G_1), \quad V^2 = F(T_2 + (1 - \gamma)T_0, R_2 - G_2).$$

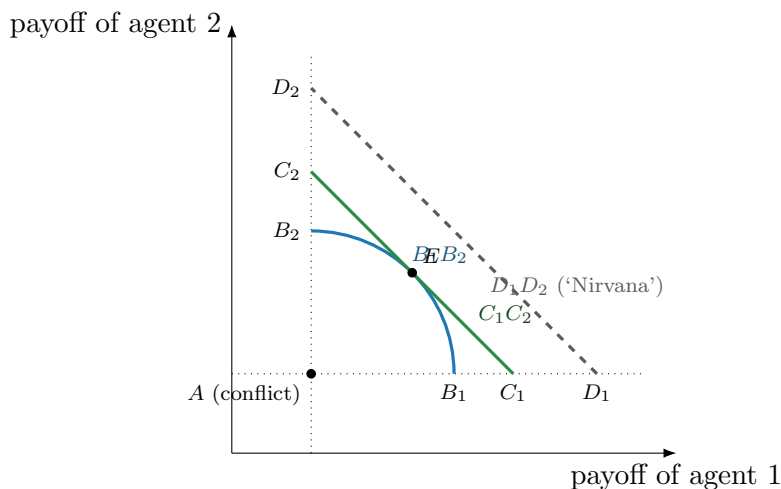
Under **Exchange**, division of land is followed by *ex post* efficient trade in factor services. With identical CRS technologies, efficiency requires all factors to be combined at the aggregate land–labor ratio — equivalently, the agents pool inputs, produce total output $F(T, L)$ with $T = T_0 + T_1 + T_2$ and $L = L_1 + L_2$, and split it with Nash-bargained share

$$\lambda(G_1, G_2) = \frac{1}{2} \left[1 + \frac{U^1 - U^2}{F(T, R_1 - G_1 + R_2 - G_2)} \right] :$$

the agent with the better war outside option gets more than half.

The key contrast: what is contestable?

Under Limited Settlement the contest is over the disputed land T_0 *only*. Under Exchange, because relative bargaining power (via $U^1 - U^2$) prices the *entire* output, the contest effectively extends over the **whole surplus** $F(T, L)$. A bigger negotiable pie means a bigger marginal return to guns.



For *given* guns: A is the conflict point; B_1B_2 the frontier from dividing land only; C_1C_2 the (linear, by CRS) frontier when division is followed by exchange, tangent to B_1B_2 at the no-further-gains allocation E ; and D_1D_2 the neoclassical secure-property frontier that would obtain with *zero* guns — the model's limiting case is exactly the Edgeworth box.

15.3 The main proposition

Proposition 1 (Skaperdas–Syropoulos 2002)

With identical secure endowments ($R_1 = R_2$, $T_1 = T_2$) and Nash-bargained sharing rules in both regimes, in any symmetric interior equilibria each agent **(i) produces fewer guns under Limited Settlement than under Exchange**, and **(ii) obtains a higher payoff under Limited Settlement than under Exchange**. *Proof idea.* The appendix shows that at any symmetric gun profile the proportional marginal return to arming is strictly larger under Exchange,

$$\frac{1}{V^1} \frac{\partial V^1}{\partial G_1} \Big|_{G_1=G_2} < \frac{1}{W^1} \frac{\partial W^1}{\partial G_1} \Big|_{G_1=G_2},$$

because a marginal gun improves the threat point by the same amount in both regimes, but under Exchange that improvement is leveraged over the *whole* output (C_1C_2 dominates B_1B_2). More guns dissipate more labor; with identical agents there are no *ex post* gains from trade to compensate; hence Exchange is Pareto-dominated. *Ex post efficiency causes ex ante inefficiency.*

Asymmetric endowments: when Exchange wins after all

With $F = T^\alpha L^{1-\alpha}$ and the ratio-form CSF, numerical analysis (the paper's Fig. 2) shows Exchange is the *ex ante* Pareto-dominant regime when: (i) endowments are sufficiently **dis-similar** (large gains from trade); (ii) the contested resource is sufficiently **important** in production (α large — then Limited Settlement's contest over land is nearly as big as Exchange's contest over output, so the arming differential shrinks); and (iii) overall **insecurity is small**. Otherwise Limited Settlement dominates or is preferred by one agent — and since it takes two to trade, Limited Settlement is then the stable outcome. **Interpretation:** secure property and free exchange are complements; where insecurity is rampant (feudal Europe, mercantilist state rivalry), restrictions on exchange — even on *secure* final goods produced with insecure inputs — can be individually and collectively rational.

Understanding Quiz 15 — Lesson 15

Q15.1 Given guns, dividing the contested resource in proportion to winning probabilities dominates fighting because:

- (A) fighting is illegal
- (B) F is strictly concave in land, so the sure share beats the lottery with the same mean
- (C) guns are refundable
- (D) the CSF is symmetric

Q15.2 The deep reason Exchange induces more arming than Limited Settlement is that:

- (A) exchange requires more transport
- (B) under Exchange the effective contest extends over the whole output, not just the disputed land, so the marginal return to improving one's threat point is larger
- (C) bargaining is impossible under Exchange
- (D) guns are cheaper under Exchange

Q15.3 With identical agents, Proposition 1 implies:

- (A) Exchange Pareto-dominates because gains from trade are maximal

- (B) Limited Settlement Pareto-dominates: it induces less arming, and identical agents have no ex post gains from trade to lose
- (C) both regimes are equivalent
- (D) conflict is the unique equilibrium

Q15.4 Exchange is more likely to be *ex ante* Pareto dominant when:

- (A) endowments are dissimilar, the contested resource is important in production, and insecurity is low
- (B) endowments are identical
- (C) insecurity is maximal
- (D) labor is the dominant input

16 Lesson 16: Trade and Insecure Resources — Garfinkel–Skaperdas–Syropoulos (2015)

★ Garfinkel, M.R., S. Skaperdas and C. Syropoulos (2015), “Trade and Insecure Resources,” *Journal of International Economics*, 95(1), 98–114

Setting. The full general-equilibrium development of Lesson 14: two small countries contest a resource (“oil”) used in producing tradable goods, arm to enforce claims, and can operate under different trade regimes; world prices are parametric. Because trade determines product prices and hence *relative factor prices*, it alters the marginal cost of converting resources into military force: **security policies are trade-regime dependent**. **Key results.**

- **Price threshold logic (generalizing Lesson 14).** Arming under free trade is driven by the world relative price of the good intensive in the contested resource. When that price is *high* relative to autarky, the prize is more valuable and/or guns are effectively cheap: opening to trade *amplifies* the dispute. When it is low, trade *pacifies* it. Autarky Pareto-dominates free trade over an intermediate price range, exactly as in the $[\tilde{p}, \tilde{p}^A]$ interval of Lesson 14.
- **What is traded matters less than what is contested.** Because trade in goods embodies factor content, restricting trade only in the contested input is not enough: free exchange of *secure* final goods produced with insecure inputs transmits the same security externality.
- **Welfare is non-monotone in openness.** The gains-from-trade triangle competes with the arming (enforcement-cost) margin; either can dominate, so trade liberalization between adversaries is not automatically welfare-improving — but neither is protection.
- **Security-motivated trade policy** again requires precommitment: a trade regime chosen before arming can improve welfare by disarming the rival.

16.1 The mechanism in one equation

The envelope decomposition of Lesson 14 carries over verbatim: for country i facing world price p of the labor-intensive good,

$$\frac{d\tilde{V}_i^F}{dp} = \Gamma p^{-\beta} \left[-M_i^F + T_0 \phi_{G_j} \frac{dG_j^F}{dp} \right],$$

a **terms-of-trade effect** plus a **security externality**. All of the paper's regime comparisons are signs of these two terms in different price regions; the contribution is to show the logic survives general neoclassical (Heckscher–Ohlin flavored) production, asymmetric countries, and alternative specifications of which goods and factors can cross borders.

Companion: Garfinkel, Skaperdas and Syropoulos (2008), “Globalization and Domestic Conflict,” *Journal of International Economics*, 76(2), 296–308

The same apparatus turned *inward*: two groups within a single small open economy contest a domestic resource. Trade openness fixes the price of the contested-resource-intensive good at world levels. **Result:** opening to trade *intensifies domestic conflict* when the world price of what the contested resource produces exceeds its autarky level (typically: natural-resource exporters), and pacifies it otherwise; a commodity-price boom is a conflict shock. This is the theoretical backbone for the empirical literature linking world commodity prices to civil war intensity (the “resource curse” in its conflict form), and the domestic mirror image of Lesson 14's realist region.

Understanding Quiz 16 — Lesson 16

Q16.1 In GSS (2015), security policies are “trade-regime dependent” because:

- (A) armies require imported equipment
- (B) the trade regime determines product and hence relative factor prices, which set the marginal cost of producing military force
- (C) defense is a public good
- (D) the WTO regulates arming

Q16.2 Restricting trade only in the contested input is insufficient because:

- (A) smuggling is inevitable
- (B) goods trade embodies factor content: exchanging secure final goods made with insecure inputs transmits the same security externality
- (C) the contested input is never traded
- (D) final goods are always secure

Q16.3 In GSS (2008), globalization intensifies *domestic* conflict when:

- (A) the country imports the contested-resource-intensive good
- (B) the world price of the good produced with the contested resource exceeds its autarky level, raising the value of the prize
- (C) tariffs are high
- (D) groups are identical

Q16.4 The welfare comparison of trade regimes between adversaries is:

- (A) always in favor of free trade, by the gains-from-trade theorem
- (B) non-monotone: gains from trade compete with induced changes in arming, and either can dominate

- (C) always in favor of autarky
- (D) independent of world prices

17 Lesson 17: Arming in the Global Economy — Garfinkel–Syropoulos–Yotov (2020)

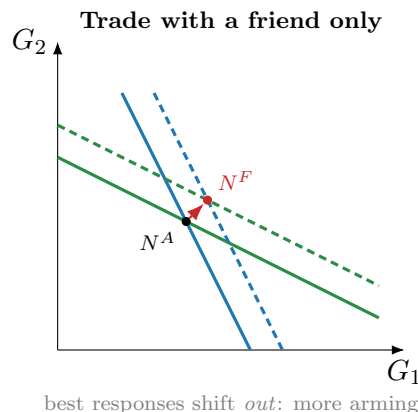
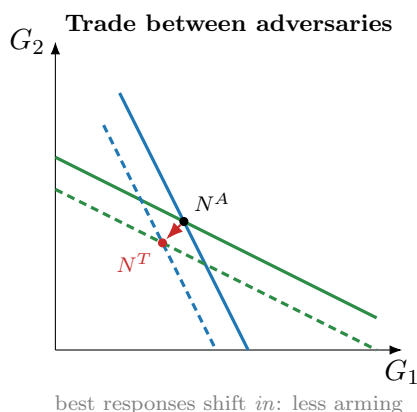
★ Garfinkel, M.R., C. Syropoulos and Y.V. Yotov (2020), “Arming in the Global Economy: The Importance of Trade with Enemies and Friends,” *Journal of International Economics*, 123, 103295

Setting. Lessons 14 and 16 shut down the terms-of-trade channel by assuming small countries. Here the adversaries are *large*: two countries contest a productive resource and may trade with each other and/or with a third, *friendly* country; arming moves world prices. The trade regime therefore changes not only the opportunity cost of guns but the terms-of-trade consequences of using them. **Key results.**

- **Trade between adversaries is pacifying.** Trade between the two enemies reduces each one’s incentive to arm *given the opponent’s arming* (best responses shift in). With sufficiently similar mixes of initial endowments, both arm less, resources wasted on conflict shrink, and both enjoy the gains from trade: openness is doubly welfare-improving.
- **Asymmetries temper, but do not overturn, the aggregate result.** With sufficiently different endowment mixes, a move to trade can induce *one* adversary to arm more, so trade need not be welfare-improving for both — but **aggregate arming falls**.
- **Trade with friends only is conflict-enhancing.** If the adversaries do not trade with each other but each trades with the third (friendly) country, a move from autarky to trade *intensifies* their conflict: both arm more, and the extra arming can be costly enough to swamp the gains from trade.
- **Empirics.** Structural-gravity evidence on trade costs, alliances, and military expenditure is consistent with the theory’s comparative statics.

17.1 The terms-of-trade discipline mechanism

Why does trading *with* the enemy discipline arming? Arming enlarges a country’s share of the contested resource and hence its production of the resource-intensive good. When the adversaries trade with each other, dumping more of that good on their common market *depresses its relative price* — a terms-of-trade loss that the aggressor itself bears. Each gun thus carries an extra, internalized price penalty, exactly the flip side of Lesson 2’s optimum-tariff logic: market power turns quantity expansion into self-inflicted price damage. When instead each adversary trades only with a *friend*, capturing more of the resource *improves* the country’s terms of trade vis-à-vis the friend — the world market happily absorbs the loot at a good price — so the effective prize is bigger and best responses shift *out*.



Reading the result against the liberal–realist debate

The paper splits the classical liberal claim in two. *Interdependence between the disputants* genuinely pacifies (a terms-of-trade discipline, operative only when the adversaries actually trade with each other). *Openness in general* does not: integration with third parties can finance and inflame the bilateral dispute. “Trade with enemies” and “trade with friends” are different policies with opposite security externalities — a sharp guide for interpreting sanctions, decoupling, and friend-shoring, and the large-country completion of Lesson 14’s small-country price thresholds.

Understanding Quiz 17 — Lesson 17

Q17.1 In GSY (2020), trade *between* the two adversaries:

- (A) raises each one’s incentive to arm given the rival’s arming
- (B) lowers each one’s incentive to arm given the rival’s arming: extra guns depress the price of the resource-intensive good the aggressor sells, a self-inflicted terms-of-trade loss
- (C) leaves arming unchanged
- (D) is assumed impossible

Q17.2 When the two adversaries trade only with a third, friendly country:

- (A) conflict intensifies: both arm more, and welfare can fall below autarky despite the gains from trade
- (B) conflict disappears
- (C) aggregate arming falls
- (D) the friend arms instead

Q17.3 With sufficiently *different* endowment mixes, a move to trade between adversaries:

- (A) always benefits both
- (B) can induce one country to arm more and lose, but aggregate arming still falls
- (C) causes war
- (D) has no effect on arming

Q17.4 The channel that makes security policies trade-regime dependent here, and that was absent in Lessons 14 and 16, is:

- (A) the opportunity-cost-of-guns channel
- (B) the terms-of-trade channel: the adversaries are large, so arming moves world prices

- (C) the revenue channel
- (D) the exchange-rate channel

18 Lesson 18: War, Peace, and Commitment — Garfinkel–Syropoulos (2019, 2021)

Lesson 15 proved that, *given guns*, settlement Pareto-dominates war. Why, then, do wars happen at all in complete-information conflict models — and what does it take for peace to be *self-enforcing*? Two papers supply the answer: war can dominate settlement *ex ante* because of commitment problems (2019), and unarmed peace is sustainable only under specific configurations of destruction, insecurity, and distribution (2021).

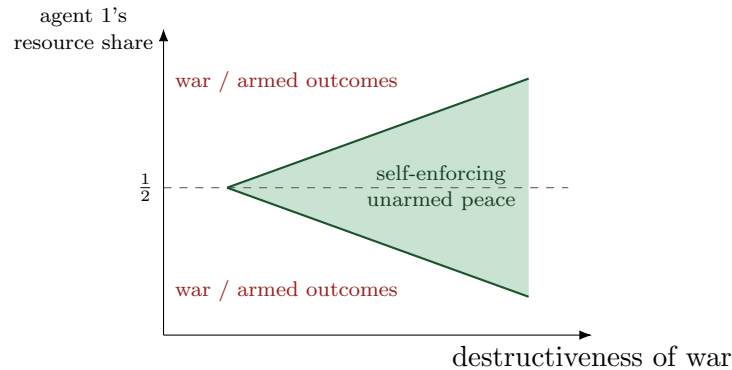
Garfinkel, M.R. and C. Syropoulos (2019), “Problems of Commitment in Arming and War: How Insecurity and Destruction Matter,” *Public Choice*, 178(3–4), 349–369

Agents cannot commit to future arming: a settlement today must be defended by guns again tomorrow, whereas a decisive war today — though destructive — can *lock in* control and eliminate future arming costs. War is thus a (costly) **commitment device**. **Result:** open conflict is chosen in equilibrium when war is not too destructive and property is sufficiently insecure; higher destructiveness and greater security both enlarge the domain of peaceful settlement. The Pareto ranking of Lesson 15 is a *given-guns*, one-shot statement; once the future stream of enforcement costs is priced in, rational agents can prefer the gamble of war.

★ Garfinkel, M.R. and C. Syropoulos (2021), “Self-Enforcing Peace Agreements that Preserve the Status Quo,” *Games and Economic Behavior*, 130, 148–178

Setting. A single-period, guns-versus-butter, complete-information model in which two agents dispute an *insecure share* of their combined output. The choice is between **unarmed peace** — an agreement that preserves the status quo division with *no* arming at all — and **war**, which is possibly destructive. Equilibrium requires immunity to both *unilateral* and *coalitional* deviations. **Key results.**

- Peace can, but need not, emerge: the outcome depends on **war’s destructiveness**, the **degree of output security**, and the **initial distribution of resources**.
- For unarmed peace to be stable at *every* feasible endowment distribution, war must be **destructive**: the savings on guns alone are not enough of a peace dividend.
- This contrasts sharply with **armed peace** modeled as bargaining over the contested pie (Garfinkel–Syropoulos 2018 and Lesson 15): there, given guns, division always beats war, so peace-as-bargaining is always an equilibrium possibility. Peace that must dispense with guns *and* freeze the status quo is much harder to sustain.
- **Ex ante transfers without commitment can buy peace** — a resource-rich agent voluntarily cedes resources to relax the poorer rival’s predation constraint — but only for some configurations of security and destruction parameters.



The wedge captures the paper's comparative statics: peace preserving the status quo needs destructive war to deter deviation, and survives a wider range of resource distributions the more destructive war is; highly uneven distributions invite predation by the poorer or seizure by the stronger, collapsing unarmed peace.

Understanding Quiz 18 — Lesson 18

Q18.1 In GS (2019), war can rationally be chosen over settlement, despite Lesson 15's dominance result, because:

- (A) agents are irrational
- (B) settlement must be re-defended by costly arming every period, while a decisive war locks in control and eliminates future arming costs — war is a costly commitment device
- (C) war is never destructive
- (D) information is incomplete

Q18.2 GS (2021) show that unarmed peace preserving the status quo is stable at every endowment distribution only if:

- (A) output is fully insecure
- (B) war is destructive — the savings on guns alone are an insufficient peace dividend
- (C) agents are identical
- (D) transfers are prohibited

Q18.3 The equilibrium concept in GS (2021) requires immunity to:

- (A) unilateral deviations only
- (B) both unilateral and coalitional deviations
- (C) trembling-hand perturbations
- (D) renegotiation only

Q18.4 Ex ante resource transfers without commitment:

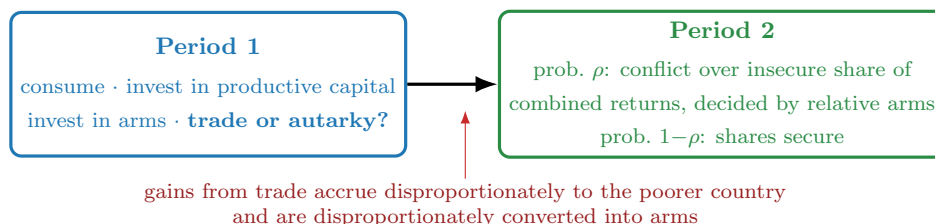
- (A) never help
- (B) can improve the prospects for peace, but only for some configurations of output security and war's destructiveness
- (C) always guarantee peace
- (D) are equivalent to arming

19 Lesson 19: Prudence versus Predation — Garfinkel–Syropoulos–Zylkin (2022)

★ Garfinkel, M.R., C. Syropoulos and T. Zylkin (2022), “Prudence versus Predation and the Gains from Trade,” *Journal of Economic Theory*, 201

Setting. A *dynamic*, two-country model. In period 1 each country splits its resources among current consumption, **productive investment** (future productive capacity), and **arming** (future military capacity), and the countries decide whether to trade with each other today. In period 2, with positive probability, conflict breaks out over a portion of the combined returns to first-period investments, resolved by the CSF applied to accumulated arms. The question: when do the effects of *current trade* on the *future balance of power* make trade unappealing? **Key results.**

- **Prudence vs. predation.** A positive probability of future conflict induces the country with *less* resource wealth to “prey” on the more “prudent” behavior of its richer rival — tilting its portfolio toward guns — and the more so, the likelier is conflict.
- **Trade shifts the balance of power.** A move from autarky to trade raises *both* countries’ current incomes, but the **poorer country realizes the relatively larger income gain** and devotes a **relatively larger share of that gain to arming**. Today’s mutual gains from trade thus finance tomorrow’s adverse shift in relative military capacity from the rich country’s viewpoint.
- **Endogenous refusal to trade.** The richer country rationally chooses *autarky* when the initial wealth gap is sufficiently large, and is more likely to prefer autarky the higher the probability of future conflict — a fully rational, security-based rejection of mutually profitable exchange.
- **Evidence.** The period surrounding the end of the Cold War provides suggestive empirical support: the collapse in the perceived probability of conflict coincided with the opening of East–West trade.



Why this is the modern theory of export controls and decoupling

Lesson 14’s security externality was static: trade changes today’s price of guns. Here the externality is *intertemporal and distributional*: trade changes who can afford guns tomorrow. The result rationalizes strategic trade restrictions against rising rivals — refusing gains from trade today to avoid financing an adverse shift in the future balance of power — without any appeal to irrationality, misperception, or lobbying. Note the inversion of Grossman–Helpman (Lesson 13): there, protection was a political failure of a welfare-maximizing benchmark;

here, protection against a potential adversary can *be* the welfare-maximizing benchmark.

Understanding Quiz 19 — Lesson 19

Q19.1 In GSZ (2022), a positive probability of future conflict induces:

- (A) the richer country to arm relatively more
- (B) the poorer country to “prey” on the richer rival’s prudence by tilting its portfolio toward arms, more so as conflict becomes more likely
- (C) both countries to disarm
- (D) immediate war

Q19.2 Although trade raises both countries’ current incomes, the richer country may refuse it because:

- (A) the gains from trade are negative for large countries
- (B) the poorer country gains relatively more and converts a relatively larger share of its gain into arming, shifting the future balance of power against the rich country
- (C) tariff revenue is lost
- (D) consumers prefer domestic goods

Q19.3 The richer country is more likely to choose autarky when:

- (A) the wealth gap is small and conflict is unlikely
- (B) the initial wealth difference is large and the probability of future conflict is high
- (C) both countries are poor
- (D) arms are ineffective

Q19.4 The empirical episode the paper uses as suggestive evidence is:

- (A) the Napoleonic blockade
- (B) the period surrounding the end of the Cold War
- (C) the 1930s tariff wars
- (D) Bretton Woods

20 Lesson 20: The Frontier — War-and-Peace Dynamics, Technology, and Sanctions

★ Garfinkel, M.R. and C. Syropoulos (2025), “Trade Interdependence, Arming and the Choice between War and Peace,” Working Paper (CESifo 11802)

Setting. The synthesis of Lessons 18 and 19: a dynamic model in which two countries with competing claims to a resource first **arm**, then choose **war** or **peaceful settlement**. War precludes international trade and can be destructive, but *locks in* gains and eliminates all future arming costs; peace — possibly supported by arming — avoids destruction and allows mutually advantageous trade, but every future settlement must be *renegotiated under the threat of war*. **Key results.** The paper characterizes when peace is stable: depending on war’s destructiveness, time preferences, and the distribution of endowments, **greater gains from trade can pacify international tensions — but possibly only for sufficiently uneven endowment distributions**. Arming that supports a negotiated division *different* from the status quo expands the scope for peace relative to the unarmed status-quo agreements of Lesson 18: armed peace is easier to sustain than disarmament.

★ Camacho, O., M.R. Garfinkel, C. Syropoulos and Y.V. Yotov (2025), “Geopolitical Frictions and Technology Transfers: Theory and Empirics,” Working Paper

Question. How do geopolitical frictions shape the diffusion of technology? **Setting.** A guns-versus-butter model with a technology *leader* and a *laggard*; transfers of **dual-use** (general-purpose) technology raise the laggard’s productivity — and therefore both its gains from exchange with the leader and its capacity to arm. **Key results.** When the initial technological distance is large, output security is low, and the laggard’s absorptive capacity is limited, the leader *blocks* the transfer: the security externality (a better-armed rival) outweighs the economic gains. The analysis also uncovers a possible **low-technology trap** — blocked transfers keep the laggard far behind, which keeps transfers blocked. **Empirics.** Cross-border patent flows (proxying transfers) and sanctions (proxying geopolitical conflict) over 1995–2018 support the theory. This is Lesson 19’s logic applied to the technology margin: export controls on chips are prudence against predation.

Felbermayr, G., T.C. Morgan, C. Syropoulos and Y.V. Yotov (2023), “Towards Rationalizing the Use of Economic Sanctions,” Work in Progress

The program’s policy capstone: if trade restrictions are extensions of security policy (Lessons 14–19), then **sanctions** should be modeled as instruments in an ongoing contest — chosen by a sender to shift a target’s arming, resources, or behavior — rather than as exogenous trade-cost shocks. The project builds on the authors’ Global Sanctions Data Base and their survey work (*European Economic Review*, 2021) to ask when sanctions are imposed, of what type, and when they “work,” rationalizing observed sanction design with the conflict-theoretic apparatus of this Part. As work in progress, treat its conclusions as provisional; its role in the course is to mark where the literature is heading.

20.1 Synthesis: one framework, seven lessons

Paper	Margin of choice	Headline lesson
SS 2001 (L14)	guns vs. butter, small open	Trade levels the playing field for arming; autarky beats trade for intermediate world prices
SS 2002 (L15)	scope of settlement	Ex post efficient exchange can be ex ante Pareto-dominated: bigger negotiable pie, more arming
GSS 2008/2015 (L16)	trade regime, GE	Security policy is trade-regime dependent; price booms in contested sectors inflame conflict
GSY 2020 (L17)	who trades with whom	Trade with enemies pacifies (TOT discipline); trade with friends only inflames
GS 2019/2021 (L18)	war vs. settlement	War as commitment device; unarmed status-quo peace needs destructive war
GSZ 2022 (L19)	trade today vs. power tomorrow	Rich rationally refuse trade that arms a rising rival
GS 2025, Camacho et al. 2025, Felbermayr et al. 2023 (L20)	dynamics, technology, sanctions	Armed peace easier than disarmament; tech transfers blocked at large distance; sanctions as security policy

Every entry is one force wearing different clothes: **anarchy makes the returns to appropriation endogenous to openness**. The gains from trade are real, but so is the security externality; which dominates is a matter of prices, distributions, destruction, and time — not of ideology. The classical liberal and realist views are both special cases.

Understanding Quiz 20 — Lesson 20

Q20.1 In GS (2025), war — despite destroying trade — has the compensating features that it:

- (A) is costless
- (B) locks in gains and eliminates all future arming costs, whereas peace must be renegotiated under threat every period
- (C) improves the terms of trade
- (D) is reversible

Q20.2 In Camacho et al. (2025), the technology leader blocks a dual-use transfer when:

- (A) the laggard is close behind and property is secure
- (B) technological distance is large, output security is low, and the laggard's absorptive capacity is limited
- (C) patents are strong
- (D) sanctions are in place

Q20.3 The “low-technology trap” refers to:

- (A) obsolete weapons
- (B) a self-reinforcing outcome in which blocked transfers keep the laggard far behind, which

in turn keeps transfers blocked

- (C) slow patent offices
- (D) declining research productivity

Q20.4 The Felbermayr et al. project reframes sanctions as:

- (A) exogenous trade-cost shocks
- (B) endogenous instruments in an ongoing interstate contest — security policy conducted by trade means
- (C) purely humanitarian tools
- (D) revenue devices

Final Exam Supplement — Part VII (Trade, Conflict, and Security)

F17. In Skaperdas–Syropoulos (2001), opening to trade “levels the playing field” because:

- (A) endowments are equalized
- (B) both adversaries face the same world price, hence the same opportunity cost of guns
- (C) the CSF becomes symmetric
- (D) guns are traded

F18. The realist case against trade holds in the 2001 model when the world price of butter is:

- (A) above the autarky price
- (B) in an intermediate range below the autarky price, where cheap guns and a valuable prize make the security externality dominate
- (C) zero
- (D) infinite

F19. Skaperdas–Syropoulos (2002): with identical agents, ex post efficient Exchange is ex ante Pareto-dominated by Limited Settlement because:

- (A) exchange is taxed
- (B) rendering the whole output negotiable raises the marginal return to arming, and identical agents have no offsetting gains from trade
- (C) bargaining fails
- (D) land is abundant

F20. The historical correlation the 2002 paper explains is between:

- (A) war and inflation
- (B) the security of property rights and the removal of restrictions on exchange
- (C) tariffs and growth
- (D) feudalism and trade surpluses

F21. GSS (2008): a boom in the world price of a contested resource’s output:

- (A) pacifies domestic conflict
- (B) intensifies domestic conflict in countries exporting that output — the conflict face of the resource curse
- (C) has no effect
- (D) lowers arming everywhere

F22. GSY (2020): the pacifying force of trade between adversaries operates through:

- (A) alliance treaties
- (B) a terms-of-trade discipline: extra arming depresses the price of the resource-intensive good the aggressor sells

- (C) cultural exchange
- (D) fixed exchange rates

F23. GSY (2020): when adversaries trade only with a friendly third country:

- (A) arming falls
- (B) conflict intensifies and the extra arming can outweigh the gains from trade
- (C) war becomes impossible
- (D) the friend mediates

F24. GS (2021): unarmed peace preserving the status quo across all endowment distributions requires:

- (A) perfect security
- (B) destructive war — deterrence by the cost of the alternative, not merely savings on guns
- (C) side payments
- (D) incomplete information

F25. GSZ (2022): the country that rationally refuses mutually profitable trade is:

- (A) the poorer one
- (B) the richer one, when the wealth gap is large or future conflict is likely, because trade disproportionately arms the rival
- (C) the smaller one
- (D) neither, ever

F26. Camacho et al. (2025): dual-use technology transfers are blocked by the leader when:

- (A) the laggard has caught up
- (B) technological distance is large, security is low, and absorption is limited — risking a low-technology trap
- (C) patents expire
- (D) trade costs are high

Lesson quizzes — Part VII

Lesson 14 — Q14.1: B (only T_0 , m , p ; trade equalizes the price of guns). Q14.2: B (interval $(\tilde{p}, \tilde{p}^A)$). Q14.3: B ($M = 0$ at $\tilde{p}^A$; pure security externality). Q14.4: B (precommitment before arming).

Lesson 15 — Q15.1: B (concavity in land). Q15.2: B (whole output becomes negotiable). Q15.3: B. Q15.4: A (dissimilar endowments, important resource, low insecurity).

Lesson 16 — Q16.1: B (prices set the cost of military force). Q16.2: B (factor content of trade). Q16.3: B. Q16.4: B (non-monotone).

Lesson 17 — Q17.1: B. Q17.2: A. Q17.3: B (aggregate arming falls). Q17.4: B (terms-of-trade channel).

Lesson 18 — Q18.1: B (war as commitment device). Q18.2: B. Q18.3: B (unilateral and coalitional). Q18.4: B.

Lesson 19 — Q19.1: B. Q19.2: B. Q19.3: B. Q19.4: B (end of the Cold War).

Lesson 20 — Q20.1: B. Q20.2: B. Q20.3: B. Q20.4: B.

Final exam supplement — Part VII

F17: B. Same p , same opportunity cost of guns. **F18: B.** $p \in (\tilde{p}, \tilde{p}^A)$. **F19: B.** Bigger negotiable pie $\Rightarrow$ more arming; no ex post gains to offset. **F20: B.** Secure property and free exchange are complements. **F21: B.** The prize appreciates. **F22: B.** Internalized price penalty on arming. **F23: B.** Friend-only trade raises the value of the prize. **F24: B.** Deterrence requires destruction. **F25: B.** Prudence refuses to finance predation. **F26: B.** Security externality of dual-use diffusion.